Klaus Tschira Stiftung

GESELLSCHAFT FÜR INFORMATIK

de RSE GESELLSCHAFT FÜR FORSCHUNGSSOFTWARE

RSE Award

Evaluation und Gütekriterien von Forschungssoftware

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Theorie

1.1) Eure Papervorträge (und Diskussion)

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1.1) Eure Papervorträge

Title	Creator	Year
EVERSE Reference Framework	Chue Hong et al.	2026
LiCoEval: Evaluating LLMs on License Compliance in Code Generation	Xu et al.	2025
Foundational Competencies and Responsibilities of a Research Software Engineer: Current State and Suggestions for Future Directions	Goth et al.	2025
GI- und DE-RSE Muster-Leitlinie zur effizienten Entwicklung von Forschungssoftware	Czerniak et al.	2025
Towards a quality indicator for research data publications and research software publications – a vision from the Helmholtz Association	Zu Castell et al.	2024
Multi-Dimensional Categorization of Research Software with Examples	Hasselbring et al.	2024
Introducing the FAIR Principles for research software	Barker et al.	2022
From FAIR research data toward FAIR and open research software	Hasselbring et al.	2020
Good enough practices in scientific computing	Wilson et al.	2017
Ten simple rules for making research software more robust	Taschuk and Wilson	2017
Software citation principles	Smith et al.	2016
Best Practices for Scientific Computing	Wilson et al.	2014
Open Science: One Term, Five Schools of Thought	Fecher and Friesike	2014
everse.software/indicators/website/indicators.html		
Measuring What Matters in the Age of AI: The Research Software Metrics Landscape	Malviya-Thakur et al.	

1.2) Kategorisierung von Forschungssoftware: EVERSE 3-tier model



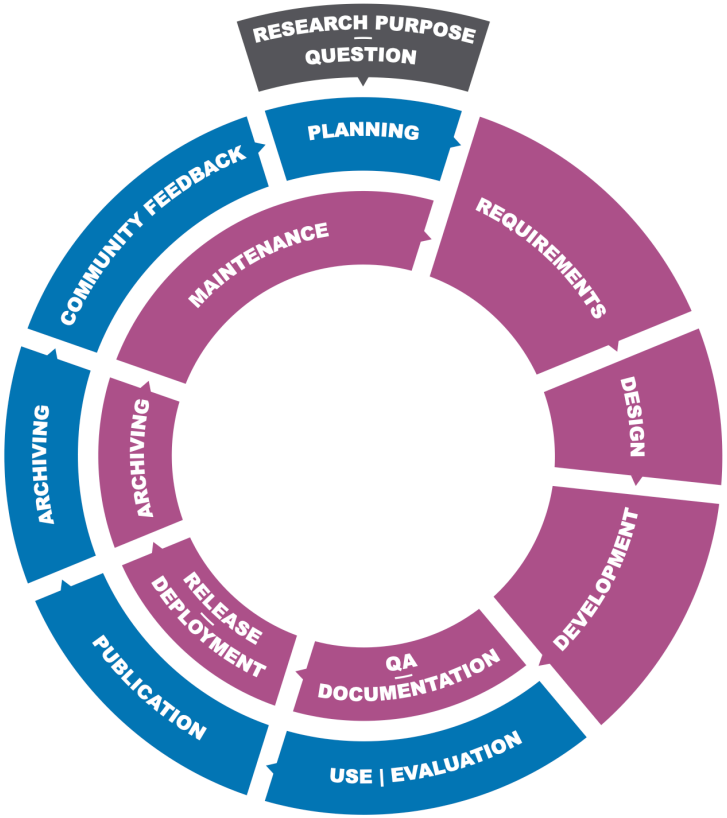
Quelle

1.2.1) EVERSE 3-tier model

Tier	Definition	Use Cases	Features	Examples	Quality Implications
Analysis code	computational research processes and methodology	simulation, data generation, preparation, analysis and visualisation	personal use, scripts, small scope	Discrete representation in neural models	functionality, reproducability, basic documentation
Prototype tools	grew beyond the project in which it originated	answer multiple research questions	developed and used by multiple people	nipype: neuroimage python library	maintainability, testing, community interaction
Research software infrastructure	broadly applicable research software	bigger researcher community	large and possibly distributed development team	Jupyter Notebooks/Jupyter Lab, SciPy, NumPy, TensorFlow	security, scalability, sustainability

Quelle

1.2.2) Research Software Lifecycle



Quelle

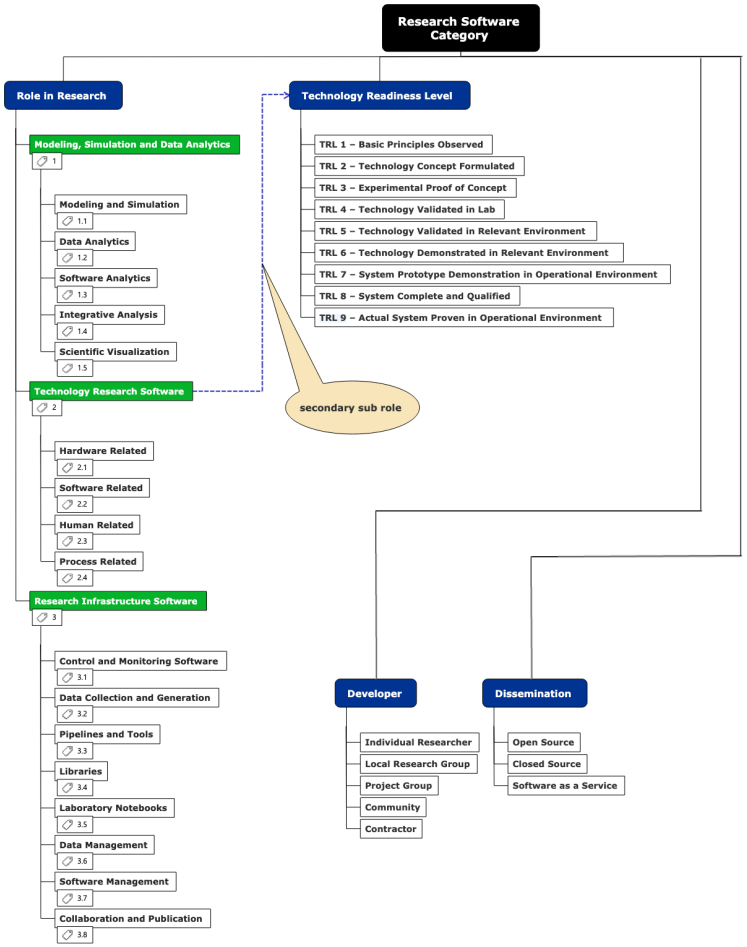
1.2.3) Quality Implications Across the Lifecycle

Different lifecycle stages emphasise different quality dimensions:

Lifecycle Phase	**Quality Focus**
Planning / Design	functional suitability, compatibility requirements
Development	maintainability, testability, security practices
Release / Deployment	reliability, performance efficiency, FAIR principles
Maintenance / Evolution	sustainability, community engagement

Quelle: EVERSE Reference Framework, Version 2.1

1.2.4) Kategorisierung nach Hasselbring et al. (2024)



1.3) Evaluation von Forschungssoftware

1. Welche Dimensionen und Kriterien für die Evaluation?
2. Wie vergleicht man Forschungssoftware aus verschiedenen Tiers und disziplinenübergreifend?
3. ...

Praxis